

ESP32-S3 Gaming console

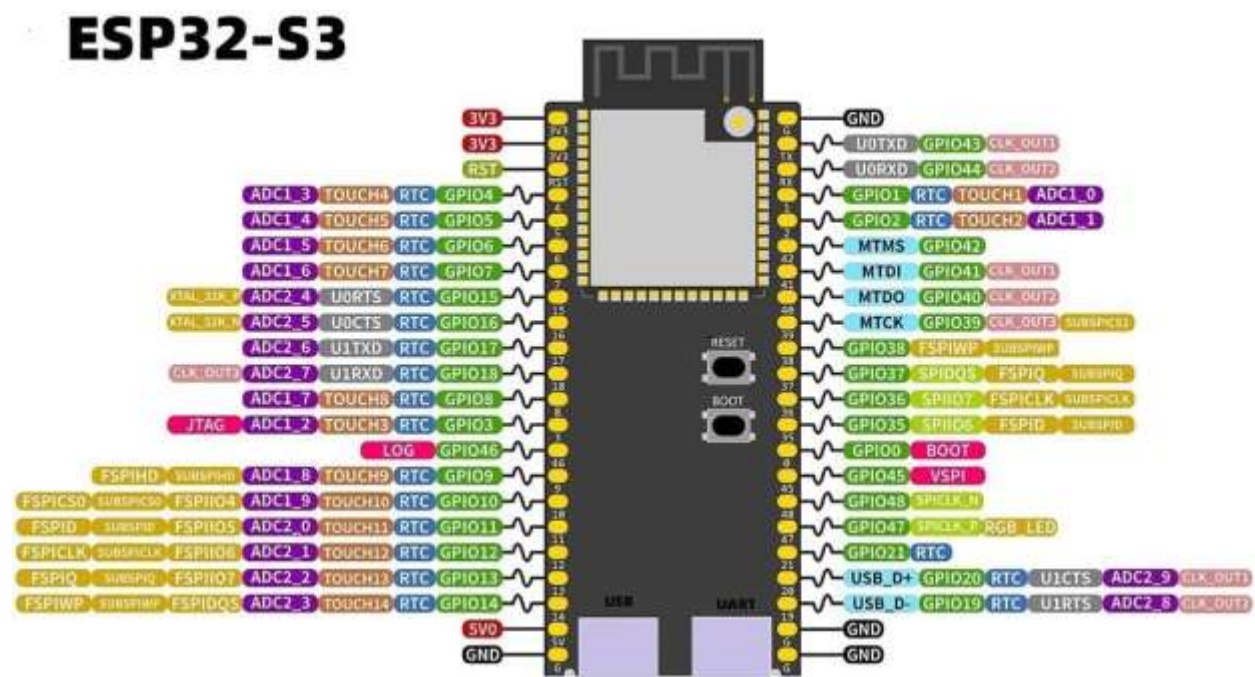
part list:

CPU

ESP32-S3 Development Board N16R8 MCU with Dual-Mode Wi-Fi Bluetooth Type-C, Compatible with Arduino IoT ESP32-S3-WROOM-1

Link:

<https://a.co/d/04EmQYCw>



Introduction to ESP32S3

Built in 512KB SRAM.

External 16MB Flash and 8MB PSRAM, suitable for complex tasks and data caching.

GPIO: Up to 45 general-purpose pins.

ADC: 14 12 bit ADC channels.

3 UARTs, supporting serial communication.

SPI/I2C/I2S: Supports common peripheral communication protocols.

PWM: Suitable for motor control or LED dimming.

USB: Supports native USB 2.0, OTG and CDC functions.

RGB controller: Built in LCD and LED controller.

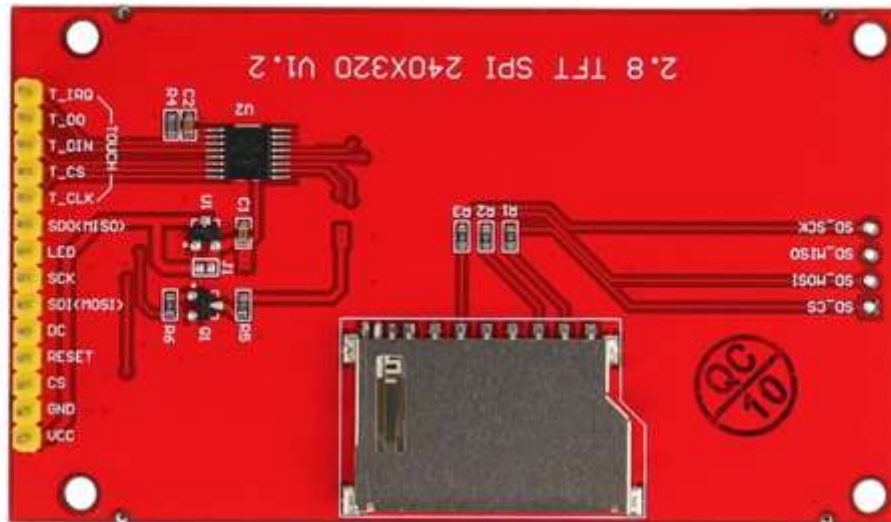


Screen

2.8" SPI TFT LCD Display ILI - 9341 Resisive Touch Screen Module | DIYmalls 2.8" SPI TFT LCD Display ILI - 9341 Resisive Touchscreen Module 320x240 5V for Arduino 2560 ESP32 Development Board

Link:

<https://a.co/d/04pHWV3C>



No.	Pin	Description
1	VCC	5V/3.3V power input
2	GND	Ground
3	CS	LCD chip select signal, low level enable
4	RESET	LCD reset signal, low level reset
5	DC	LCD register / data selection signal, high level: register, low level: data
6	SDI (MOSI)	SPI bus write data signal
7	SCK	SPI bus clock signal
8	LED	Backlight control, high level lighting, if not controlled, connect 3.3V always bright
9	SD0 (MISO)	SPI bus read data signal, pls don't connect it if don't need read function
(The following is touchscreen signal line wiring, if don't need touch function or module doesn't have touch function, you can not connect them)		
10	T_CLK	Touch SPI bus clock signal
11	T_CS	Touch screen chip select signal, low level enable
12	T_DIN	Touch SPI bus input
13	T_D0	Touch SPI bus output
14	T_IRQ	Touchscreen interrupt signal, low level when touch detected

SD card reader

**WWZMDiB 6 Pcs Micro SD TF Card Adapter Mini Reader Module 3.3V 6 Pin SPI Interface
Compatible with for Arduino Raspberry Pi ESP32**

Link:

<https://a.co/d/097Mflco>

Sound card:

**MAX98357 MAX98357A I2S 3W Class D Amplifier Breakout Interface I2S DAC Decoder
for Audio Raspberry Pi Esp32 Arduino Zero**

Link:

<https://a.co/d/0aGFAuXz>

Speaker:

**Watt 8 Ohm Speaker Mini Speaker Micro 8ohm 3w Loundspeaker with JST-PH2.0
Interface for Arduino Small Electronic Projects**

Link:

<https://a.co/d/09YtihPS>

Joystick

**Joystick Module Game Console Dual-Axis XY Joystick Module KY-023 Control Stick
Sensor Board Compatible with PS2 Arduino Raspberry Pi**

Link:

<https://a.co/d/01CceGgZ>

Power Supply

DC-DC Buck Converter Adjustable Power Module 24V to 12V 9V 5V 3V

link:

<https://a.co/d/0dv5x0FC>

Pin Layout

Display	ESP32-S3 Pin
CS	10
RESET	8
DC	9
MOSI	11
SCK	12
MISO	Disconnected
VCC	External regulated 5 V
LED	ESP32 3.3 V
GND	Shared ground

Buttons

pin 38 — X button — GND

pin 39 — Y button — GND

pin 40 — A button — GND

pin 41 — B button — GND

Joystick pin

Joystick pin	ESP32-S3
GND	Common ground
+5V	3.3V
VRX	pin 4
VRY	pin 5
SW	pin 6

SD module	ESP32-S3
3.3V	3.3V
GND	Common ground
CS	PIN 14
MOSI	PIN 11
CLK	PIN 12
MISO	PIN 13

MAX98357A	Connection
VIN	External regulated 5 V
GND	Common ground
LRC	PIN 18
BCLK	PIN 17
DIN	PIN 16
GAIN	Leave disconnected
SU/SD	Leave disconnected initially

Special thanks to

<https://www.fesliyanstudios.com/royalty-free-music/downloads-c/suspenseful-music/25>

<https://sfxr.me/>

for the sound effects and music tracks